Appendix C - Geographe Marine Park Data Analysis Report

South-west Marine Parks Network

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Data Analysis

Statistical methods and model selection results underpinning the natural values reported in Appendix A.

Benthic ecosystem extent

Table C.1: Habitat model selection results. Top generalised additive models (GAMs) for predicting the probability of occurrence of key habitats from full subset analyses. Differences between the lowest reported corrected Akaike Information Criterion (ΔAICc), AICc weight (ωAICc), variance explained (R^2) and effective degrees of freedom (EDF) are reported for model comparison. Model selection was based on the most parsimonious model (fewest variables - shown in bold) within two units of the lowest AICc.

Response	Model	Δ AICc	ω AICc	R^2	EDF
Macroalgae	**detrended+roughness+Z+year**	**0**	**0.573**	**0.08565**	**11.86**
	detrended+roughness+Z+year	**0.6**	**0.425**	**0.08504**	**10.18**
Sand	**detrended+roughness+Z+year**	**0**	**0.641**	**0.38782**	**11.64**
	detrended+roughness+Z+year	**1.176**	**0.356**	**0.38761**	**10.58**
Seagrass	**detrended+aspect+Z+year**	**0**	**0.987**	**0.42307**	**13.37**
Rock	**detrended+roughness+Z+year**	**0**	**0.721**	**0.48828**	**10.57**
Sessile invertebrates	**roughness+aspect+Z+year**	**0**	**0.168**	**0.08775**	**8.15**
	roughness+aspect+Z+year	**0.214**	**0.151**	**0.087**	**7.15**
	roughness+aspect+Z+year	**0.347**	**0.141**	**0.08749**	**7.67**
	roughness+aspect+Z+year	**0.354**	**0.141**	**0.0857**	**6.01**
	roughness+aspect+Z+year	**0.747**	**0.116**	**0.08776**	**8.49**
	roughness+aspect+Z+year	**1.008**	**0.101**	**0.08832**	**9.29**
	roughness+aspect+Z+year	**1.419**	**0.083**	**0.08626**	**7.06**
	roughness+aspect+Z+year	**1.587**	**0.076**	**0.0878**	**8.56**
Reef	**detrended+roughness+Z+year**	**0**	**0.515**	**0.0666**	**12.27**
	detrended+roughness+Z+year	**0.135**	**0.482**	**0.0661**	**10.46**

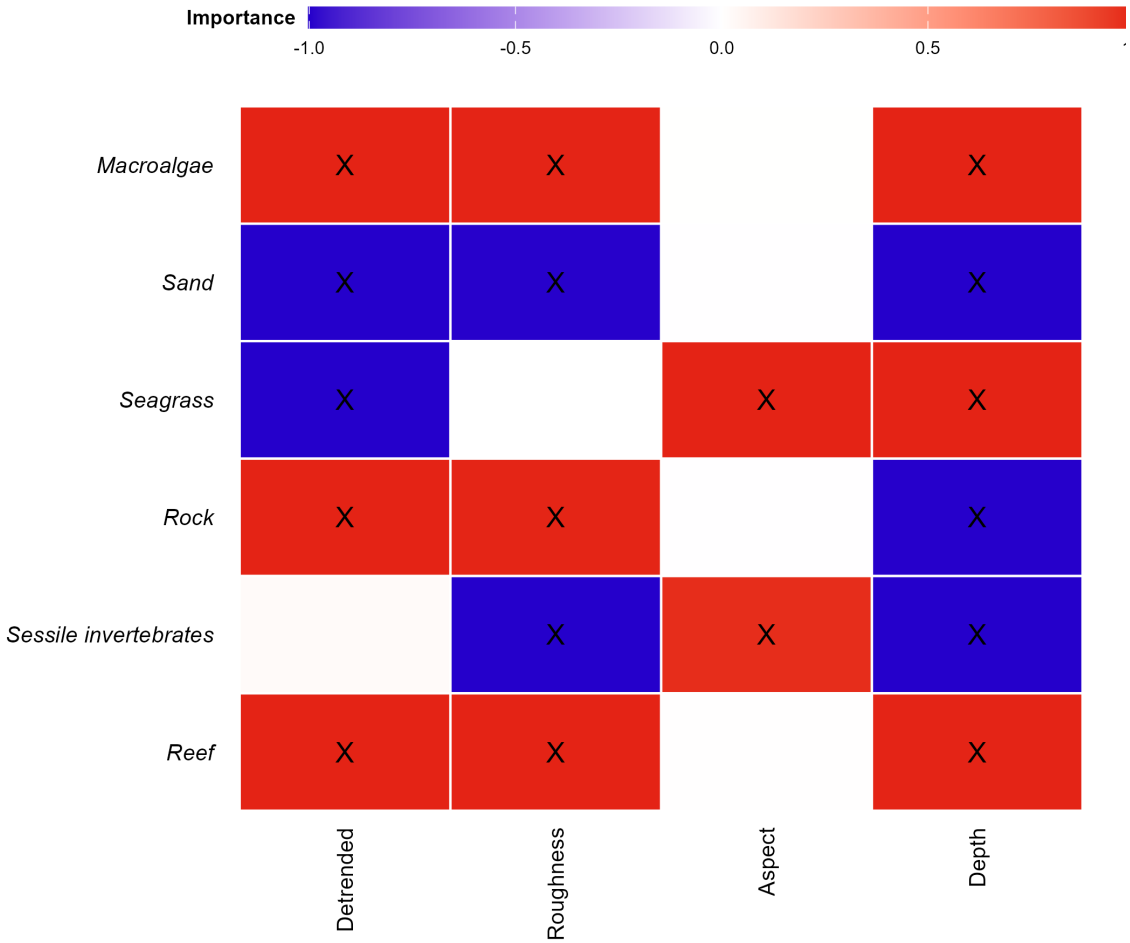


Figure C.1: Variable importance scores for predicting the probability of occurrence of main habitat categories. Response variables included in the most parsimonious top model are indicated (X), with the colour gradient representing positive (red), zero (white) and negative (blue) relationships. Aspect is a circular variable, so the direction of its relationship should be read from the response curves rather than the colour gradient.

... the per-year figure chunks ...

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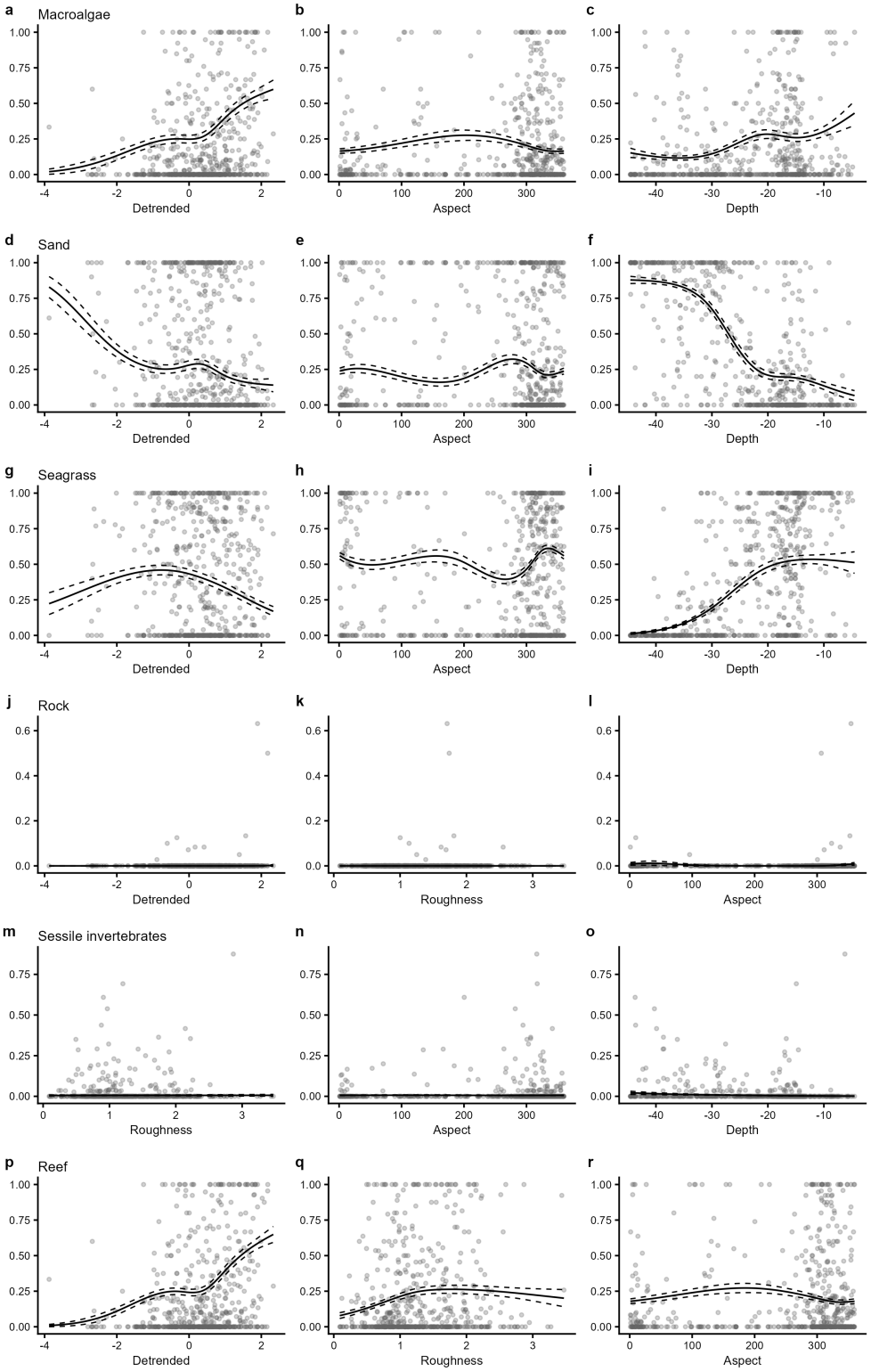


Figure C.3: Plots of the most parsimonious model found to predict the probability of occurrence of key habitat types. Individual panes display the predicted relationships between model covariates and the probability of occurrence of each habitat class. Solid lines represent predicted Generalized Additive Model (GAM) fits with other variables held constant at their mean value. Dashed lines represent upper and lower standard error bounds around the prediction, and points represent the observed data from all survey years.

Demersal fish assemblage

Table C.2: Demersal fish model selection results. Top generalised additive models (GAMs) for predicting the species and abundance distribution of metrics of interest from full subset analyses. Differences between the lowest reported corrected Akaike Information Criterion (ΔAICc), AICc weight (ωAICc), variance explained (R^2) and effective degrees of freedom (EDF) are reported for model comparison. Model selection was based on the best model (fewest variables - shown in bold) within two units of the lowest AICc. *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

Response	Model	Δ AICc	ω AICc	AICc	R^2	EDF
Species richness	detrended+aspect+Z+reef+year+status	0	0.568	3162.33	0.41115	8.70
	detrended+Z+reef+year+status	**0.548**	**0.432**	**3162.88**	**0.40897**	**7.68**
Total abundance	Z+reef+year+status	0	0.184	6511.70	0.16787	6.58
	aspect+Z+reef+year+status	0	0.184	6511.70	0.16787	7.58
	roughness+reef+year+status	1.56	0.084	6513.26	0.16279	5.74
	roughness+aspect+reef+year+status	1.56	0.084	6513.26	0.16279	6.74
	reef+year+status	**1.633**	**0.081**	**6513.34**	**0.15949**	**4.74**
	aspect+reef+year+status	1.633	0.081	6513.34	0.15949	5.74
	detrended+Z+reef+year+status	1.663	0.08	6513.37	0.16899	7.74
	detrended+aspect+Z+reef+year+status	1.663	0.08	6513.37	0.16899	8.74
Large Reef Fish Index*	detrended+aspect+Z+reef+year+status	0	0.338	7536.19	0.24629	9.29
	aspect+Z+reef+year+status	0.567	0.254	7536.76	0.23996	7.64
	detrended+Z+reef+year+status	**0.631**	**0.246**	**7536.82**	**0.24005**	**7.96**
Reef Fish Thermal Index	detrended+aspect+Z+reef+year+status	0	0.508	1373.44	0.09649	9.24
	detrended+Z+reef+year+status	**0.865**	**0.33**	**1374.30**	**0.09199**	**8.23**

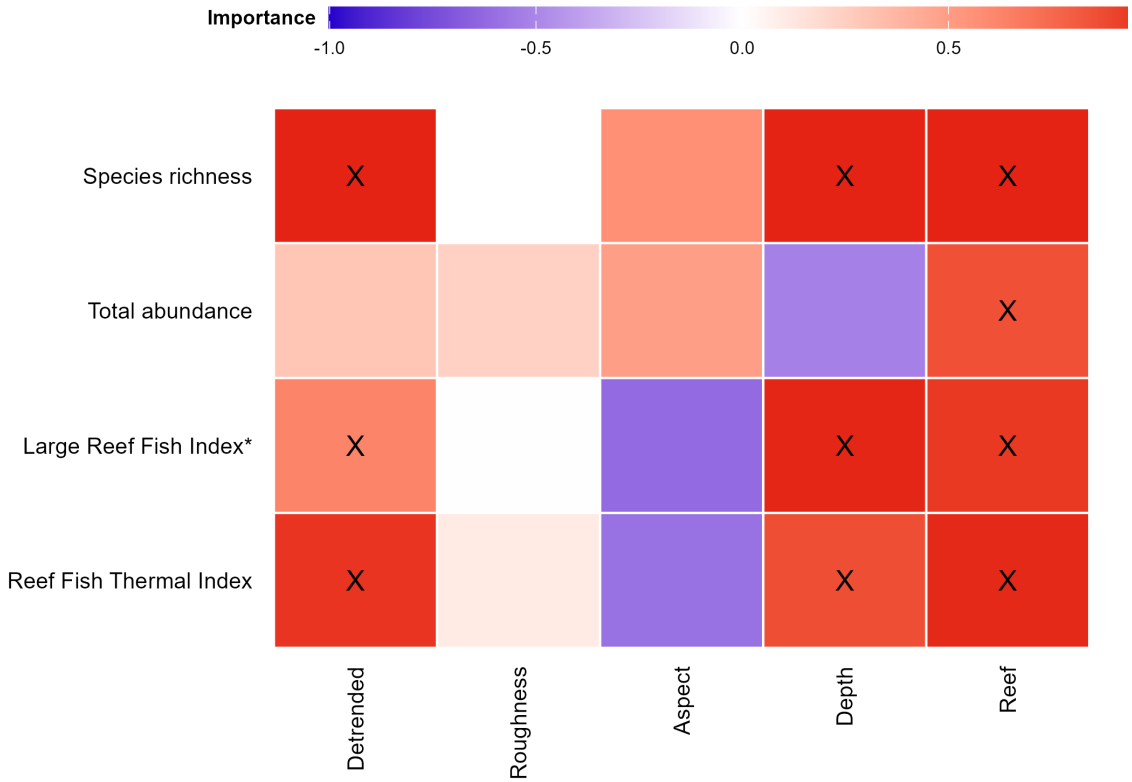


Figure C.4: Variable importance scores for predicting the probability of demersal fish metrics. Response variables included in the most parsimonious top model are indicated (X), with the colour gradient representing positive (red), zero (white) and negative (blue) relationships. Aspect is a circular variable, so the direction of its relationship should be read from the response curves rather than the colour gradient. *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

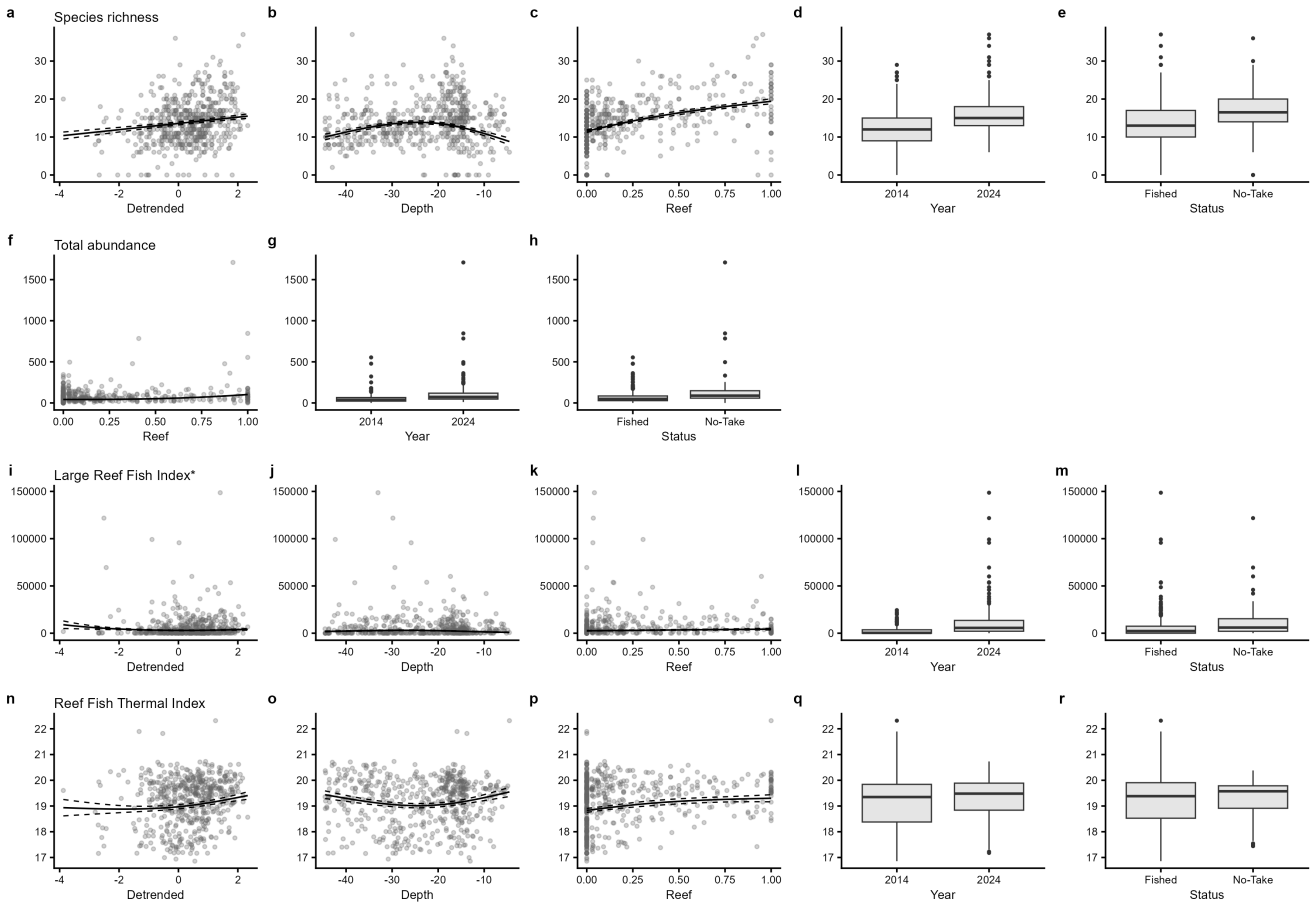


Figure C.5: Plots of the best model found to predict the probability of occurrence of key fish groups. Individual panes display the predicted relationships between model covariates and each demersal fish metric. Solid lines represent predicted Generalized Additive Model (GAM) fits with other variables held constant at their mean value, with year held at 2014 and zone status held at areas open to fishing. Dashed lines represent upper and lower standard error bounds around the prediction, and points represent the observed data from all survey years and zones. *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

Significance of year and zone status

Table C.3: Significance of the year and zone status terms in each demersal fish GAMM. Estimate, standard error (SE), test statistic and p-value are taken from the parametric coefficient table of each model summary, one row per factor-level contrast (e.g. a survey year against the reference year, or No-Take against Fished).

Response	Term	Estimate	SE	Statistic	p-value
Species richness	Zone status: No-Take vs Fished	3.749	0.689	5.442	<0.001
	Year: 2024 vs 2014	3.093	0.417	7.419	<0.001
Total abundance	Zone status: No-Take vs Fished	0.722	0.012	59.409	<0.001
	Year: 2024 vs 2014	0.609	0.011	55.891	<0.001
Large Reef Fish Index*	Zone status: No-Take vs Fished	0.512	0.191	2.681	0.008
	Year: 2024 vs 2014	1.368	0.130	10.510	<0.001
Reef Fish Thermal Index	Zone status: No-Take vs Fished	-0.024	0.139	-0.173	0.863
	Year: 2024 vs 2014	0.134	0.083	1.616	0.107

Table C.4: Zone status test for each demersal fish metric, laid out by survey year. Zone status (No-Take vs Fished) is fitted as a single term across all years (no year by status interaction), so the status estimate, SE and p-value are the same in every row for a given metric; predicted values and their SE are shown per year, with other model covariates held at their mean.

Response	Year	Fished (predicted)	Fished (SE)	No-Take (predicted)	No-Take (SE)	Status estimate	Status SE	Status p-value
Species richness	2014	13.845	0.397	17.594	0.817	3.749	0.689	<0.001
	2024	16.938	0.409	20.687	0.724	3.749	0.689	<0.001
Total abundance	2014	42.801	0.486	88.086	1.452	0.722	0.012	<0.001
	2024	78.716	0.712	162.001	1.952	0.722	0.012	<0.001
Large Reef Fish Index*	2014	2965.884	379.273	4948.306	1178.643	0.512	0.191	0.008
	2024	11646.509	1322.529	19431.135	3851.694	0.512	0.191	0.008
Reef Fish Thermal Index	2014	19.014	0.082	18.990	0.167	-0.024	0.139	0.863
	2024	19.148	0.084	19.124	0.148	-0.024	0.139	0.863